1. TENTANG AUTO-EDA DASHBOARD

Auto-EDA Dashboard (DS Generator) adalah platform analisis data otomatis berbasis web yang dikembangkan oleh **Kelompok 2 Data Science ITSB**. Aplikasi ini dirancang untuk membantu analis data, manajer, dan pengambil keputusan bisnis dalam melakukan: (1) unggah dan validasi dataset, (2) audit kesehatan data (quality audit), (3) pembersihan data interaktif (data cleaning), (4) perhitungan statistik deskriptif dan lanjutan, (5) visualisasi pola data secara interaktif, serta (6) perumusan insight dan rekomendasi strategis berbasis data secara otomatis dan real-time.

2. ANGGOTA TIM PENGEMBANG (KELOMPOK 2 ITSB)

Nama Lengkap	NIM	Peran / Fokus Analisis
Carol Dupino Pereira	52250051	Descriptive & Advanced Statistics Engine
Refantanur Husnul Haqib	52250052	Visualizations & Dynamic Plotly Dashboard
Cahaya Medina Semidang	52250053	Data Preprocessing & Sanitizer Module
Raihania Syah Putri	52250054	Time Series Forecasting & Trends Panel
Cloise Shafira	52250044	Smart Insights Generation Algorithm
Adinda Adelia Futri	52250055	Reporting System PDF/Excel & Security Sanitization

3. DESKRIPSI DATA & RINGKASAN EKSEKUTIF

Nama File Dataset:	ecommerce.xlsx
Waktu Analisis:	14 June 2026, 01:09:42
Status Kebersihan:	**RAW DATA (Perlu Pembersihan)**
Total Baris (Observasi):	1299
Total Kolom (Variabel):	22
Variabel Numerik:	8
Variabel Kategorikal:	14

Missing Cells	Duplicate Rows	IQR Outliers
0 (0.0%)	**0**	**150**

Identifikasi Masalah Kualitas Data:

- 150 outlier terdeteksi (metode IQR)
- 15 kolom memiliki inkonsistensi format teks
- 4 kolom kemungkinan tidak relevan (ID/free text)

Detail Kesehatan Per Kolom:

Kolom	Tipe	Missing (%)	Unique	Outliers	Status
Unnamed: 0	object	0 (0.0%)	1288	0	Possible ID/free text
order_date	object	0 (0.0%)	844	0	OK
ship_date	object	0 (0.0%)	871	0	OK
platform	object	0 (0.0%)	21	0	Whitespace issues
category	object	0 (0.0%)	28	0	Whitespace issues
product_name	object	0 (0.0%)	49	0	Whitespace issues
unit_price	object	0 (0.0%)	1276	0	Possible ID/free text
quantity	float64	0 (0.0%)	8	78	78 outliers
gross_sales	object	0 (0.0%)	1276	0	Possible ID/free text
campaign	object	0 (0.0%)	5	0	OK
voucher_code	object	0 (0.0%)	5	0	OK
discount_pct	float64	0 (0.0%)	5	0	OK
discount_value	object	0 (0.0%)	858	0	OK
shipping_cost	float64	0 (0.0%)	7	0	OK
net_sales	object	0 (0.0%)	1275	0	Possible ID/free text
payment_method	object	0 (0.0%)	28	0	Whitespace issues
customer_segment	object	0 (0.0%)	3	0	OK
region	object	0 (0.0%)	8	0	OK
stock_status	object	0 (0.0%)	3	0	OK
order_status	object	0 (0.0%)	12	0	Whitespace issues
customer_rating	float64	0 (0.0%)	5	72	72 outliers
priority_flag	object	0 (0.0%)	6	0	OK

3a. ALUR PROSES DATA CLEANING (PIPELINE)

Dataset diproses melalui 5 tahapan alur data cleaning di backend engine:

1. Audit Kesehatan (Data Profiling): Mendeteksi sel kosong (NaN), baris duplikat, inkonsistensi teks, dan outlier numerik (IQR).
2. Standarisasi Teks: Menghilangkan spasi berlebih dan menyeragamkan bentuk huruf (Title/Lower/Upper Case).
3. Imputasi Data Hilang: Mengisi sel kosong dengan mean/median/modus atau menghapus baris jika kerusakan ekstrem.
4. Pembersihan Outlier (IQR Capping): Menetralkan nilai ekstrim menggunakan batas IQR.
5. Drop Kolom Tidak Relevan: Menghapus kolom variansi nol atau rasio keunikan terlalu tinggi (ID/teks acak).

Metrik	Sebelum (Raw)	Sesudah (Cleaned)	Perubahan
Baris Data	2001	1299	Dihapus: 702 baris
Kolom Data	22	22	Dihapus: 0 kolom
Sel Kosong	968 (2.2%)	0 (0.0%)	Dibersihkan: 968 sel

Log Tindakan Cleaning:

• Dropped 702 rows with missing values

3b. Statistik Deskriptif — Variabel Numerik

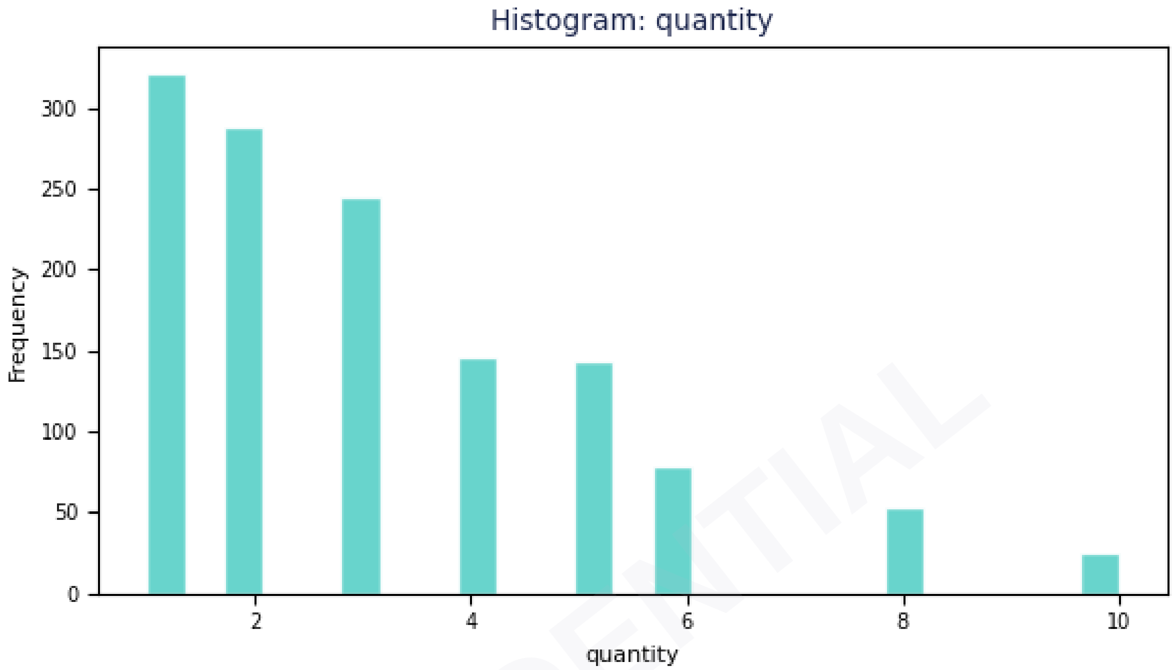
Kolom	Mean	Median	Min	Max	Std Dev	Skewness	Normality
Unnamed: 0	1011.69	1020.5	2.0	1999.0	578.87	-0.0333	Not Normal
unit_price	559112.82	395012.0	30834.0	2498385.0	507855.29	1.7643	Not Normal
quantity	3.14	3.0	1.0	10.0	2.06	1.1878	Not Normal
gross_sales	1736586.6	996528.0	31646.0	22642020.0	2227988.6	3.4332	Not Normal
discount_pct	11.37	10.0	0.0	30.0	9.7	0.3274	Not Normal
discount_value	193121.72	69598.0	0.0	6792606.0	389929.66	7.1776	Not Normal
shipping_cost	14052.35	15000.0	0.0	25000.0	7580.41	-0.4838	Not Normal
net_sales	1388968.72	794999.0	-7663533.0	16607520.0	2104831.11	2.6568	Not Normal

3c. Statistik Deskriptif — Variabel Kategorikal

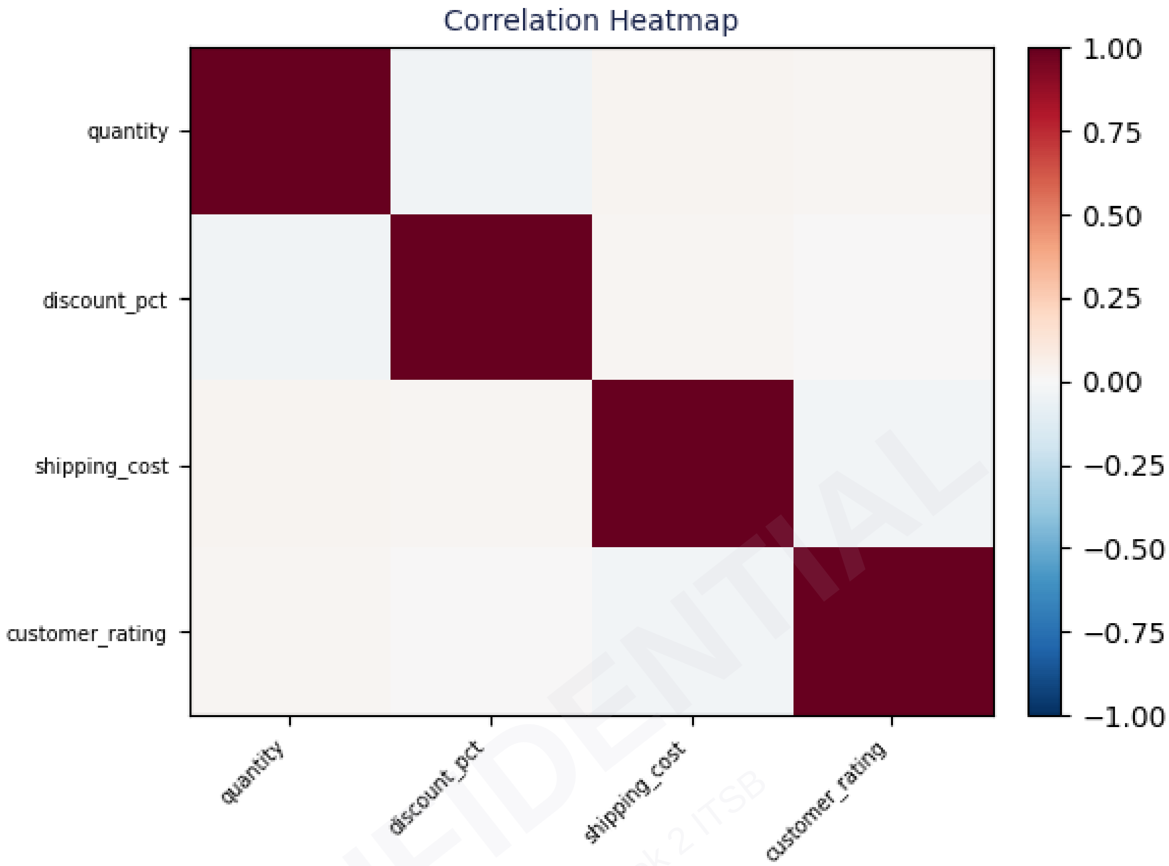
Kolom	Unique	Mode	Mode Freq	Mode %	Missing	Missing %
order_date	844	2024-09-29	5	0.38%	0	0.0%
ship_date	871	28/07/2024	5	0.38%	0	0.0%
platform	21	TikTok Shop	256	19.71%	0	0.0%
category	28	Fashion	244	18.78%	0	0.0%
product_name	49	Storage Box	62	4.77%	0	0.0%
campaign	5	Normal Day	428	32.95%	0	0.0%
voucher_code	5	NONE	428	32.95%	0	0.0%
payment_method	28	Virtual Account	256	19.71%	0	0.0%
customer_segment	3	Returning	695	53.5%	0	0.0%
region	8	Bandung	184	14.16%	0	0.0%
stock_status	3	In Stock	943	72.59%	0	0.0%
order_status	12	Delivered	305	23.48%	0	0.0%
customer_rating	5	5.0	672	51.73%	0	0.0%
priority_flag	6	No	359	27.64%	0	0.0%

4. VISUALISASI KUNCI (HISTOGRAM & HEATMAP)

4a. Histogram (Distribusi Variabel Numerik)



4b. Heatmap Korelasi (Hubungan Antar Variabel)



5. TEMUAN KUNCI & INSIGHT OTOMATIS

Excellent Data Quality — Zero Missing Values

No missing values found at all. The dataset is clean and ready for statistical analysis or machine learning without additional preprocessing.

Dataset: 1,299 Rows × 22 Columns

Dataset size is sufficiently representative for reliable statistical analysis. Consists of 8 numeric and 14 categorical columns.

Highest Average Value: gross_sales

Column **gross_sales** has the highest average value of **1,736,586.6**. Median: 996528.0, Std: 2227988.6.

Most Outliers: net_sales

Column **net_sales** has the most outliers: **124 data points (9.5%)** based on IQR method (±1.5×IQR). Check whether these outliers are data errors or valid extreme values.

Highest Std Deviation: gross_sales

Column **gross_sales** has the largest standard deviation of **2,227,988.5981** (CV=128.3%). Very high variability on this column indicates a wide data spread.

Strongest Correlation: gross_sales ↔ net_sales (r=0.886)

Very Strong (positive) correlation found between **gross_sales** and **net_sales** with r = 0.886. This relationship is highly significant and worth investigating further. (R^2 = 0.785)

Normality Test — 0/8 Columns Normal

NOT normal columns: Unnamed: 0, unit_price, quantity (+5 more). Non-normal columns (100.0%) require non-parametric tests (Mann-Whitney, Kruskal-Wallis) or data transformation (log, sqrt, Box-Cox) before parametric analysis.

Skewed Distributions: 5 Column(s)

The following columns have skewness > 1: unit_price (skew=1.76), quantity (skew=1.19), gross_sales (skew=3.43), discount_value (skew=7.18). These skewed distributions can affect parametric analysis results. Consider log or Box-Cox transformation.

Categorical Balance (order_date): Balanced

Category distribution is fairly even. Most frequent category: "2024-09-29" (0.4%). Total 844 unique categories in column order_date.

Further Analysis Recommendations

Based on the structure of this dataset, suggested techniques: One-Way ANOVA / T-Test to compare means across categorical groups; Linear / Logistic Regression for prediction and modeling; Chi-Square + Cramér's V for categorical variable association; Non-Parametric Tests (Mann-Whitney, Spearman) for non-normal columns; Principal Component Analysis (PCA) for dimensionality reduction.

6. REKOMENDASI STRATEGIS PENGAMBILAN KEPUTUSAN

R1. Investigasi Nilai Anomali (Outliers): Ditemukan data pencilan ekstrem. Disarankan tim operasional memvalidasi apakah angka tersebut valid (fluktuasi bisnis riil) atau human error. Lakukan capping/truncation sebelum pemodelan prediktif lanjutan.

R2. Transformasi Skewness Data: Kolom (unit_price, quantity, gross_sales, discount_value, net_sales) memiliki kemiringan distribusi tinggi (skewed). Sebelum menerapkan algoritma berbasis distribusi normal, lakukan transformasi logaritma atau Box-Cox untuk meminimalkan bias.

R3. Pembersihan Berkala (Data Governance): Jadwalkan proses pembersihan data secara rutin menggunakan Auto-EDA Dashboard sebelum laporan akhir bulan diekspor ke departemen eksekutif, guna menjamin keputusan dibuat berdasarkan data higienis.